

ROMAN BEAMES

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EDUCATION

M.S. in Cybersecurity Engineering

Auburn University | Auburn, AL | Expected: May 2028

B.S. in Computer Science, Summa Cum Laude | GPA: 3.9/4.0

The University of Oklahoma | Norman, OK | Graduated: May 2026

Relevant Coursework: Computer Security, Cryptography, Operating Systems, Algorithm Analysis, Compiler Construction

PROJECTS

End-To-End Computer Vision Data Pipeline w/ CVAT Annotation Platform

Jan 2026 - May 2026

- Designed a containerized data pipeline for ingesting and annotating autonomous driving data using Docker, FastAPI, PostgreSQL, MinIO, and CVAT.
- Implemented backend workflows for route ingestion, metadata extraction, job execution, and annotation management through the CVAT API.
- Integrated distributed OpenPilot and CVAT worker services for data acquisition, artifact uploads, task synchronization, and processing-status coordination.
- Deployed Linux-based services supporting secure handling of Comma AI tokens, sensitive driving data, and stored artifacts.

LLM-Based Semantic Binary Similarity Analysis

Dec 2025 - May 2026

- Built a binary analysis pipeline using Binary Ninja HLIL to assess LLM-based semantic similarity across Intel and ARM compiled C++ binaries.
- Developed Python tools for binary comparison, similarity matrix generation, and deterministic scoring using OpenAI-compatible APIs and models including gpt-oss:20b, gpt-oss:120b, and llama3.3:70b.
- Analyzed LLM behavior for nondeterminism, parsing failures, and context limitations across repeated binary comparison experiments.
- Evaluated binary similarity scoring reliability for cybersecurity-oriented binary analysis, focusing on false-positive reduction across architectures.

LEADERSHIP & INVOLVEMENT

Member

Tau Beta Pi | Mar 2025 - Present

Southwest District Vice President for Programs

Kappa Kappa Psi | Apr 2023 - Mar 2026

- Coordinated programming and organizational initiatives for chapters across five states serving 300+ student members.

Squad Leader

The Pride of Oklahoma | Jun 2024 - Jan 2026

- Mentored and trained an 18-member section within a 350-member organization while collaborating with student leadership teams.

SKILLS

Languages:

Python, Java, C++, C, C#, SQL, Bash

Dev Tools:

Git, GitHub, VS Code, Eclipse IDE, Docker, PostgreSQL, MinIO, Binary Ninja, OpenSSL, Microsoft Azure, Linux

Core Competencies:

Malware Analysis, Binary Analysis, Secure Networking, Distributed Systems, Performance Analysis